

Pontryagin-Based Solver with Universally Smoothed Hamiltonian, Adaptive ΔT , and PA-Bundle Refinement

March 15, 2026

- ① Previous Meeting
- ② Error Estimate for Symplectic Euler
Numerical Results
- ③ PA-Bundle Initialization

To do:

- Re-run computations using a consistent approach for optimal control 'u' (use the closed-form Hamiltonian for testing, then a numerical optimizer as needed).

Hamiltonian function

Inputs: (x, p, t) , candidate_controls, restricted, use_oracle, bounds = $(u_{\min}, u_{\max})$

1 **Initialize candidate set:** $\mathcal{C} \leftarrow \emptyset$, bounds $\leftarrow (u_{\min}, u_{\max})$ (if available).

2 **(IF) oracle fast-path:**

if use_oracle $\wedge \exists u^*(x, p, t)$ (valid) : $\hat{u} \leftarrow \Pi_{[u_{\min}, u_{\max}]}(u^*(x, p, t))$, $\hat{H} \leftarrow \phi(\hat{u})$, **return** $(\hat{H}, \hat{u})$.

3 **(IF) bounds available: add extremal candidates**

if $(u_{\min}, u_{\max})$ exists: $\mathcal{C} \leftarrow \mathcal{C} \cup \{u_{\min}, u_{\max}\}$ (and, in general, all 2^m corners).

4 **(IF) scalar control: add a dense grid (current fallback)**

if $m = 1$: $\mathcal{C} \leftarrow \mathcal{C} \cup \{u_{\min} + k\Delta u\}_{k=0}^{G-1}$, ($G = 203$).

5 **(FOR) include provided controls (bundle controls)**

for $u \in \text{candidate_controls}$: $u \leftarrow \Pi_{[u_{\min}, u_{\max}]}(u)$ (if bounds), $\mathcal{C} \leftarrow \mathcal{C} \cup \{u\}$.

6 **(FOR) de-duplicate (tolerance-based)**

for $u \in \mathcal{C}$: keep u only if $\min_{v \in \mathcal{C}_{\text{uniq}}} \|u - v\|_2 \geq 10^{-10}$, $\mathcal{C}_{\text{uniq}} \leftarrow \mathcal{C}_{\text{uniq}} \cup \{u\}$.

Set $\mathcal{C} \leftarrow \mathcal{C}_{\text{uniq}}$.

7 **(FOR) evaluate candidates and enforce constraints**

$\hat{H} \leftarrow +\infty$, $\hat{u} \leftarrow \emptyset$.

for $u \in \mathcal{C}$: if $f(x, u, t) \notin T_K(x)$: continue; if $u \notin A$: continue; if $\phi(u) < \hat{H}$: $(\hat{H}, \hat{u}) \leftarrow (\phi(u), u)$.

8 **Fallback (rare):** if $\hat{u} = \emptyset$, ignore viability/admissibility and re-minimize over $\mathcal{C}$.

9 **Return:** $(\hat{H}, \hat{u})$, where $\hat{H} = \min_{u \in \mathcal{C}} \phi(u)$.

Optimal Hamiltonian evaluation

① **Trigger:** if $m = 1$ (scalar control).

② **Bounds:** set $a_{\text{lo}} = u_{\text{min}}, a_{\text{hi}} = u_{\text{max}}$.

③ **Call bounded Brent (SciPy):**

$\hat{u}_{\text{Brent}} = \text{minimize_scalar}(\phi_{\text{feas}}, [a_{\text{lo}}, a_{\text{hi}}], \text{method} = \text{"bounded"})$.

④ **Accept:** if `success` and $\phi_{\text{feas}}(\hat{u}_{\text{Brent}}) < \infty$, add $\hat{u}_{\text{Brent}}$ to the candidate set:

$$\mathcal{C} \leftarrow \mathcal{C} \cup \{\hat{u}_{\text{Brent}}\}.$$

⑤ **Downstream (unchanged):** the usual candidate loop selects $\hat{H} = \min_{u \in \mathcal{C}} \phi_{\text{feas}}(u)$.

PA-Bundle

By defining

$$g_i(x, t) := \partial_p H(\hat{p}_i, x, t) \quad (1)$$

$$d_i(x, t) := H(\hat{p}_i, x, t) - g_i(x, t)\hat{p}_i \quad (2)$$

We construct a piecewise-affine (PA) surrogate

$$\bar{H}(p, x, t) = \min_{1 \leq i \leq M} \{g_i(x, t) \cdot p + d_i(x, t)\}$$

which satisfies

$$H(p, x, t) \leq \bar{H}(p, x, t)$$

Envelope Theorem: If $u^*(\hat{p}_i, x_i, t_i)$ is a minimizer for $(\hat{p}_i, x_i, t_i)$ then

$$\partial H_p(\hat{p}_i, x_i, t_i) = f(t_i, x_i, u^*(\hat{p}_i, x_i, t_i)) \quad (3)$$

Pre-PA Bundle Initialization

- 1 First PA-Bundle Surrogate:

$$\bar{H}(p, x, t) = \min_{1 \leq i \leq M} \{f(t, x, u_i) \cdot p + \ell(t, x, u_i)\}, \quad u_i \in \{u_{\min}, u_{\max}, u_{\text{mid}}\} \quad (4)$$

- 2 Run a coarse solve at the first outer loop iteration (k=0):

$$(X, P) = \text{solve}(\text{problem}, \text{nodes}, \text{bundle}, \dots)$$

- 3 Return

$$t_{0:N}, X_{0:N}, P_{0:N}$$

PA-Bundle Initialization

Inputs: $t_{0:N}$, $X_{0:N}$, $P_{0:N}$, `bundle`, `restricted`, `use_oracle`, `grid_size`

- ① **Support nodes:** pick representative indices

$$\mathcal{I} \subset \{0, \dots, N\}, \quad \mathcal{I} = \text{unique}(\text{round}(\text{linspace}(0, N, k))), \quad k = \min(\text{num_support_nodes}, N + 1).$$

- ② **(Optional) 1D fallback grid:** only if bounds exist and $m = 1$

$$\mathcal{G} = \{a_j\}_{j=1}^G \subset [u_{\min}, u_{\max}], \quad G = \text{grid_size}.$$

- ③ **For each** $i \in \mathcal{I}$: set (x_i, p_i, t_i) and define

$$\phi_i(u) := p_i^\top f(x_i, u, t_i) + \ell(x_i, u, t_i).$$

- ④ **Oracle first (fast-path):**

if `use_oracle` $\wedge \exists u^* : u_{\text{oracle}} \leftarrow u^*(x_i, p_i, t_i; \text{restricted})$, **if accepted:** `bundle.add.control( $u_{\text{oracle}}$ )` and continue

- ⑤ **Fallback (only if $\mathcal{G} \neq \emptyset$):** grid-search best scalar control

$$\hat{u} \in \arg \min_{a \in \mathcal{G}} \phi_i([a]) \quad \text{s.t. } [a] \in A \text{ and (if restricted) } f(x_i, [a], t_i) \in T_K(x_i),$$

then `bundle.add.control( $\hat{u}$ )`.

- ⑥ **Return:** `added` = number of *new* controls inserted (counted via `num_planes` before/after).

The Followign example is taken from Karlsson et. al. [1]:

Hypersensitive optimal control

Minimize

$$\int_0^{25} \left(X(t)^2 + \alpha(t)^2 \right) dt + \gamma (X(25) - 1)^2 \quad (5)$$

subject to

$$X'(t) = -X(t)^3 + \alpha(t), \quad 0 < t \leq 25, \quad (6)$$

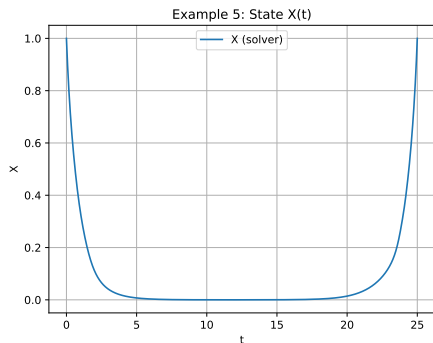
$$X(0) = 1. \quad (7)$$

for $\gamma = 1e6$. The Hamiltonian is then given by

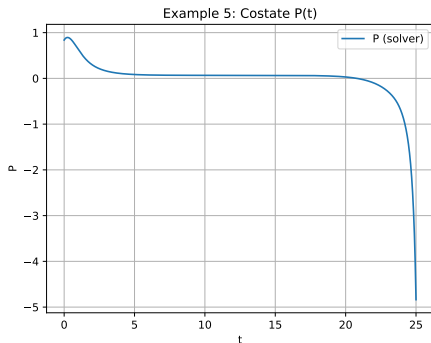
$$H(x, \lambda) = \min_{\alpha} \{ -\lambda x^3 + \lambda \alpha + x^2 + \alpha^2 \} = -\lambda x^3 - \frac{\lambda^2}{4} + x^2 \quad (8)$$

Numerical Results: Explicit Hamiltonian

We consider the same example with $s = 0.25$ and $M = 2$ (since $c \approx 1$).



(a) Solution $\bar{X}$ of the discrete optimization problem (5) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{PA} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$



(b) Solution $\bar{\lambda}$ of the discrete optimization problem (5) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{PA} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$

Numerical Results: Explicit Hamiltonian

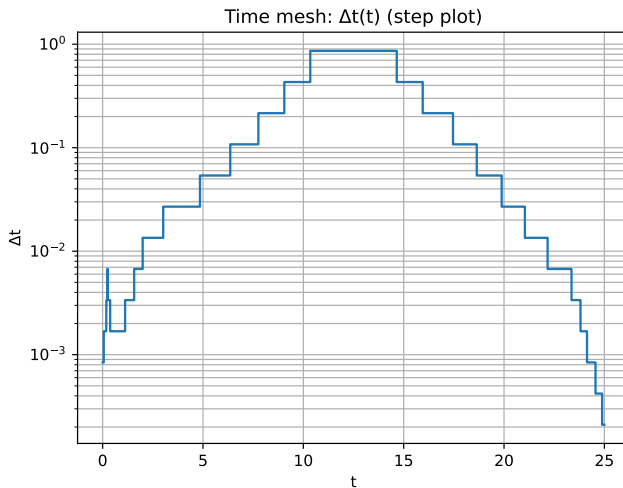
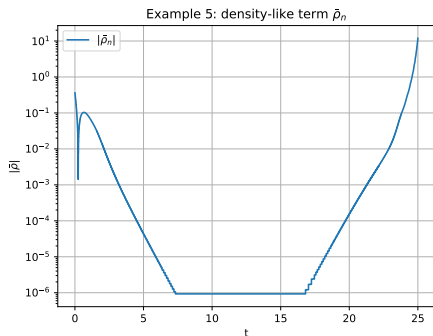
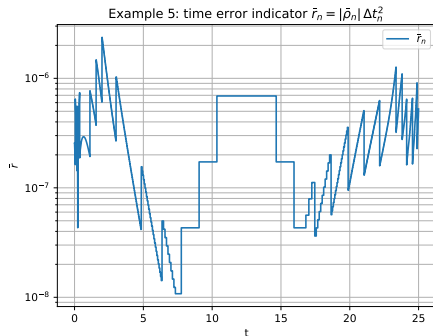


Figure: Mesh Δt for the optimization problem (5) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$

Numerical Results: Explicit Hamiltonian



(a) Error densities, $|\bar{\rho}|$ for $\varepsilon_{\text{time}} = 1e-2$,
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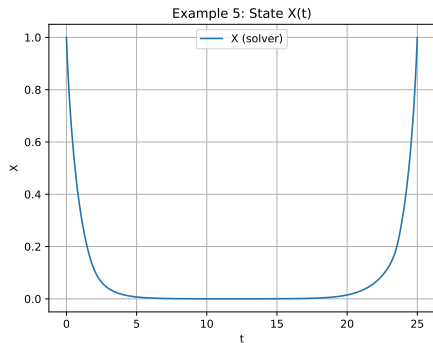


(b) Error indicators $\bar{r}_n$ of the discrete optimization problem (5) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$,
 $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$

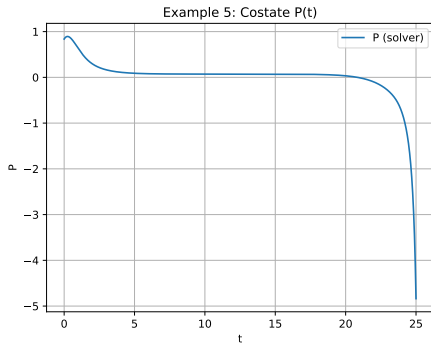
Numerical Results: Explicit Hamiltonian

#	N	M	$dt_{\min}$	$dt_{\max}$	δ	N.lt	res	η_{time}	η_{PA}	η_{δ}	action
00	29	7	8.62e-01	8.62e-01	2e-02	9	1.18e-10	2.25e-01/3.45e-04	3.43e-01/1e-02	2.39e-01/1e-02	refine_time (m=13)
01	42	7	4.31e-01	8.62e-01	2e-02	13	1.32e-10	4.77e-01/2.38e-04	6.70e-01/1e-02	2.29e-01/1e-02	refine_time (m=26)
02	68	7	2.16e-01	8.62e-01	2e-02	8	7.10e-11	2.18e-01/1.47e-04	5.85e-01/1e-02	2.29e-01/1e-02	refine_time (m=43)
03	111	7	1.08e-01	8.62e-01	2e-02	5	7.82e-11	9.91e-02/9.01e-05	5.61e-01/1e-02	2.27e-01/1e-02	refine_time (m=71)
04	182	7	5.39e-02	8.62e-01	2e-02	9	4.57e-11	4.21e-02/5.49e-05	5.95e-01/1e-02	2.26e-01/1e-02	refine_time (m=100)
05	282	7	2.69e-02	8.62e-01	2e-02	5	9.07e-11	1.13e-02/3.55e-05	5.95e-01/1e-02	2.26e-01/1e-02	refine_time (m=157)
06	439	7	1.35e-02	8.62e-01	2e-02	7	5.05e-10	2.98e-03/2.28e-05	6.01e-01/1e-02	2.26e-01/1e-02	refine_time (m=209)
07	648	7	6.73e-03	8.62e-01	2e-02	6	5.82e-10	7.61e-04/1.54e-05	6.03e-01/1e-02	2.25e-01/1e-02	refine_time (m=306)
08	954	7	3.37e-03	8.62e-01	2e-02	4	8.91e-11	1.92e-04/1.05e-05	6.03e-01/1e-02	2.26e-01/1e-02	refine_time (m=392)
09	1346	7	1.68e-03	8.62e-01	2e-02	3	1.70e-09	4.81e-05/7.43e-06	6.03e-01/1e-02	2.26e-01/1e-02	refine_time (m=537)
10	1883	7	8.42e-04	8.62e-01	2e-02	3	1.58e-10	1.20e-05/5.31e-06	6.03e-01/1e-02	2.26e-01/1e-02	refine_time (m=539)
11	2422	7	4.21e-04	8.62e-01	2e-02	3	1.17e-10	3.01e-06/4.13e-06	6.03e-01/1e-02	2.26e-01/1e-02	add_plane
12	2422	8	4.21e-04	8.62e-01	2e-02	5	1.10e-10	2.18e-06/4.13e-06	5.05e-01/1e-02	2.26e-01/1e-02	add_plane
13	2422	9	4.21e-04	8.62e-01	2e-02	4	1.44e-10	2.49e-06/4.13e-06	4.87e-01/1e-02	2.26e-01/1e-02	add_plane
14	2422	10	4.21e-04	8.62e-01	2e-02	4	2.17e-11	2.44e-06/4.13e-06	4.83e-01/1e-02	2.27e-01/1e-02	add_plane
15	2422	11	4.21e-04	8.62e-01	2e-02	5	2.93e-12	3.36e-06/4.13e-06	3.90e-01/1e-02	2.31e-01/1e-02	add_plane
16	2422	12	4.21e-04	8.62e-01	2e-02	6	4.91e-09	1.41e-05/4.13e-06	1.17e-01/1e-02	2.51e-01/1e-02	refine_time (m=680)
17	3102	12	2.10e-04	8.62e-01	2e-02	3	2.15e-10	3.54e-06/3.22e-06	1.17e-01/1e-02	2.51e-01/1e-02	refine_time (m=420)
18	3522	12	2.10e-04	8.62e-01	2e-02	2	4.97e-09	8.88e-07/2.84e-06	1.17e-01/1e-02	2.51e-01/1e-02	add_plane
19	3522	13	2.10e-04	8.62e-01	2e-02	3	1.51e-10	9.11e-07/2.84e-06	1.15e-01/1e-02	2.52e-01/1e-02	add_plane
20	3522	14	2.10e-04	8.62e-01	2e-02	3	2.08e-10	9.11e-07/2.84e-06	1.12e-01/1e-02	2.52e-01/1e-02	add_plane
21	3522	15	2.10e-04	8.62e-01	2e-02	3	1.73e-10	9.09e-07/2.84e-06	1.11e-01/1e-02	2.53e-01/1e-02	add_plane
22	3522	16	2.10e-04	8.62e-01	2e-02	4	1.01e-10	1.28e-06/2.84e-06	9.31e-02/1e-02	2.58e-01/1e-02	add_plane
23	3522	17	2.10e-04	8.62e-01	2e-02	3	2.22e-10	1.26e-06/2.84e-06	8.88e-02/1e-02	2.60e-01/1e-02	add_plane
24	3522	18	2.10e-04	8.62e-01	2e-02	5	6.42e-11	2.37e-06/2.84e-06	2.28e-02/1e-02	3.65e-01/1e-02	add_plane

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(b) Solution $\bar{\lambda}$ of the discrete optimization problem (5) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$

Numerical Results: Numeric Hamiltonian

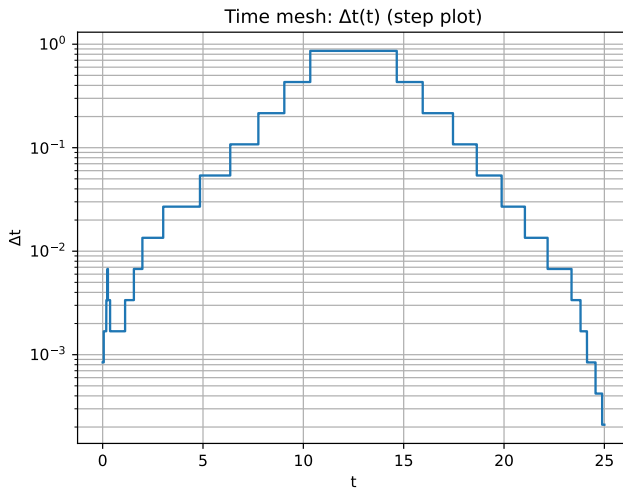
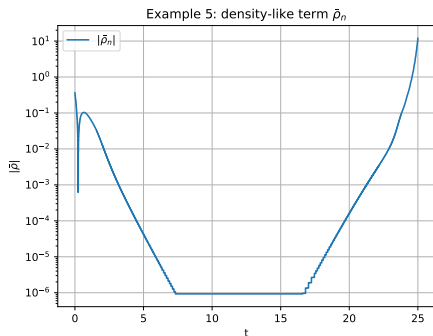
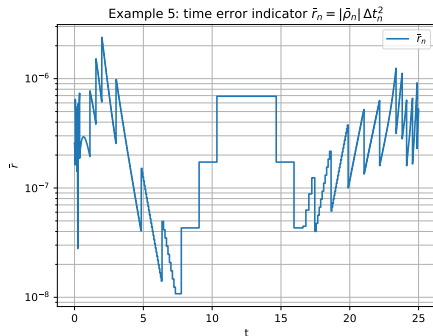


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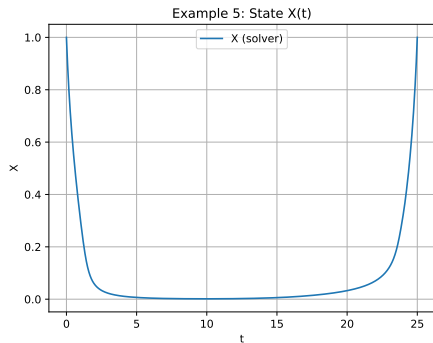
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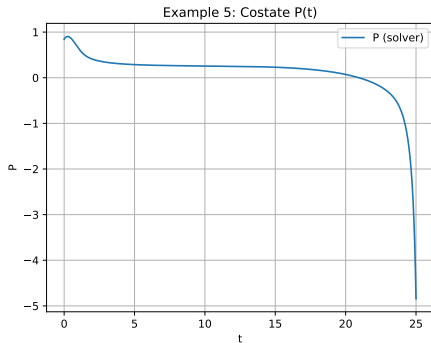
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Numerical Results: Optimal PA-Bundle Initialization

We consider the same example with $s = 0.25$ and $M = 2$ (since $c \approx 1$).



(a) Solution $\bar{X}$ of the discrete optimization problem (5) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$



(b) Solution $\bar{\lambda}$ of the discrete optimization problem (5) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$

Numerical Results: Optimal PA-Bundle Initialization

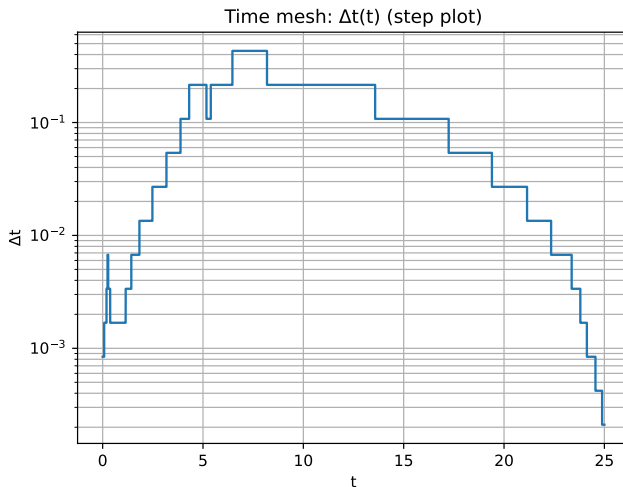
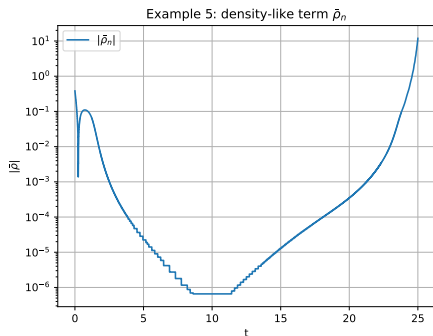
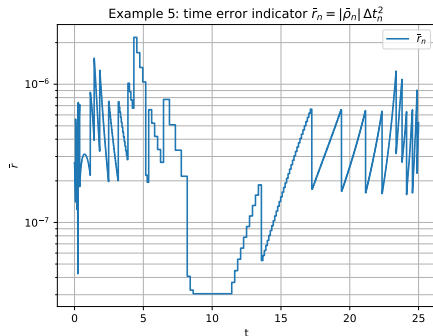


Figure: Mesh Δt for the optimization problem (5) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$

Numerical Results: Optimal PA-Bundle Initialization



(a) Error densities, $|\bar{\rho}|$ for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$



(b) Error indicators $\bar{r}_n$ of the discrete optimization problem (5) for $\varepsilon_{\text{time}} = 1e-2$, $\varepsilon_{\text{PA}} = 1e-2$, $\varepsilon_{\delta} = 1e-2$, $\delta_0 = 0.02$

Numerical Results: Optimal PA-Bundle Initialization

#	N	M	$dt_{\min}$	$dt_{\max}$	δ	N.lt	res	η_{time}	η_{PA}	η_{δ}	action
00	29	8	8.62e-01	8.62e-01	2e-02	9	1.03e-09	2.24e-01/3.45e-04	1.65e+00/1e-02	5.34e-01/1e-02	refine_time (m=10)
01	39	8	4.31e-01	8.62e-01	2e-02	13	4.61e-11	4.77e-01/2.56e-04	1.89e+00/1e-02	5.10e-01/1e-02	refine_time (m=18)
02	57	8	2.16e-01	8.62e-01	2e-02	7	2.09e-10	2.18e-01/1.75e-04	1.79e+00/1e-02	5.04e-01/1e-02	refine_time (m=31)
03	88	8	1.08e-01	8.62e-01	2e-02	5	3.22e-11	9.93e-02/1.14e-04	1.76e+00/1e-02	4.99e-01/1e-02	refine_time (m=50)
04	138	8	5.39e-02	8.62e-01	2e-02	8	1.21e-10	4.21e-02/7.25e-05	1.79e+00/1e-02	4.97e-01/1e-02	refine_time (m=83)
05	221	8	2.69e-02	8.62e-01	2e-02	5	5.45e-11	1.13e-02/4.52e-05	1.78e+00/1e-02	4.96e-01/1e-02	refine_time (m=134)
06	355	8	1.35e-02	8.62e-01	2e-02	6	2.52e-09	2.98e-03/2.82e-05	1.79e+00/1e-02	4.96e-01/1e-02	refine_time (m=185)
07	540	8	6.73e-03	8.62e-01	2e-02	6	1.35e-09	7.61e-04/1.85e-05	1.79e+00/1e-02	4.95e-01/1e-02	refine_time (m=293)
08	833	8	3.37e-03	8.62e-01	2e-02	4	1.34e-10	1.92e-04/1.20e-05	1.79e+00/1e-02	4.95e-01/1e-02	refine_time (m=381)
09	1214	8	1.68e-03	8.62e-01	2e-02	3	4.39e-09	4.81e-05/8.24e-06	1.78e+00/1e-02	4.95e-01/1e-02	refine_time (m=496)
10	1710	8	8.42e-04	8.62e-01	2e-02	3	1.81e-10	1.20e-05/5.85e-06	1.78e+00/1e-02	4.95e-01/1e-02	refine_time (m=417)
11	2127	8	4.21e-04	8.62e-01	2e-02	3	1.82e-10	3.01e-06/4.70e-06	1.78e+00/1e-02	4.95e-01/1e-02	add_plane
12	2127	9	4.21e-04	8.62e-01	2e-02	5	1.21e-10	4.72e-06/4.70e-06	1.68e+00/1e-02	4.96e-01/1e-02	refine_time (m=577)
13	2704	9	2.10e-04	8.62e-01	2e-02	3	1.17e-10	1.18e-06/3.70e-06	1.68e+00/1e-02	4.96e-01/1e-02	add_plane
14	2704	10	2.10e-04	8.62e-01	2e-02	4	1.36e-10	1.36e-06/3.70e-06	1.67e+00/1e-02	4.96e-01/1e-02	add_plane
15	2704	11	2.10e-04	8.62e-01	2e-02	4	1.89e-10	1.21e-06/3.70e-06	1.66e+00/1e-02	4.97e-01/1e-02	add_plane
16	2704	12	2.10e-04	8.62e-01	2e-02	5	8.14e-11	3.41e-06/3.70e-06	1.60e+00/1e-02	5.05e-01/1e-02	add_plane
17	2704	13	2.10e-04	8.62e-01	2e-02	10	2.00e-10	2.41e-05/3.70e-06	3.59e-01/1e-02	5.02e-01/1e-02	refine_time (m=435)
18	3139	13	2.10e-04	4.31e-01	2e-02	3	1.41e-10	7.52e-06/3.19e-06	3.58e-01/1e-02	5.02e-01/1e-02	refine_time (m=336)
19	3475	13	2.10e-04	4.31e-01	2e-02	3	9.71e-11	2.18e-06/2.88e-06	3.58e-01/1e-02	5.02e-01/1e-02	add_plane
20	3475	14	2.10e-04	4.31e-01	2e-02	3	2.40e-11	2.18e-06/2.88e-06	3.56e-01/1e-02	5.02e-01/1e-02	add_plane
21	3475	15	2.10e-04	4.31e-01	2e-02	3	1.63e-10	2.18e-06/2.88e-06	3.55e-01/1e-02	5.03e-01/1e-02	add_plane
22	3475	16	2.10e-04	4.31e-01	2e-02	3	2.87e-11	2.18e-06/2.88e-06	3.52e-01/1e-02	5.03e-01/1e-02	add_plane
23	3475	17	2.10e-04	4.31e-01	2e-02	4	1.69e-10	2.18e-06/2.88e-06	3.38e-01/1e-02	5.12e-01/1e-02	add_plane
24	3475	18	2.10e-04	4.31e-01	2e-02	3	1.95e-10	2.18e-06/2.88e-06	3.34e-01/1e-02	5.14e-01/1e-02	add_plane

Initial Planes

- ① Controls added using optimal search:

$$\mathcal{C} = \{-0.8595, -2.735 \times 10^{-7}, -4.669 \times 10^{-11}, 2.783 \times 10^{-7}, 0.1292\}. \quad (9)$$

- ② Controls added using grid with 51 nodes:

$$\mathcal{C} = \{-0.84, -0.04, 0.04, 0.12\}. \quad (10)$$

Optimal Planes Initialization

- From the previous experiments we see that a criterion for selecting good planes in the initialization of the PA-Bundle surrogate is necessary. The selected planes need to properly cover the Hamiltonian.
- Based on this example, we have ideas to evaluate scalability for multi dimensional problems.



Karlsson, J., Larsson, S., Sandberg, M., Szepessy, A., Tempone, Raul

An error estimate for symplectic euler approximation of optimal control problems.
arxiv, (2014).